

Human-Centric Robotic Control Using Speech Recognition and Vision Validation

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Abstract-- To enhance the provision of intuitive controllability of the robot, the paper proposes a human-friendly robotic manipulation system that fuses both speech-based interaction with object detection based on vision. Using essential voice recognition, such as selecting items on the basis of color, a user may control a robotic arm with the proposed system. The library SpeechRecognition Python is applied to access the Google Speech Recognition service with the help of which the microphone records the speech input and translates it into a text command. An overhead camera constantly monitors the robot workspace, and image processing methods based on the OpenCV are used to identify colored objects. In order to detect red, green and blue objects with strong color segmentation, the captured frames are converted into HSV color space. Position and centroid of the detected objects will then be obtained with the help of contour detection. A coordinate transformation will be made possible through homography to convert the image coordinates into robot workspace coordinates to allow the robot to be gingerly manipulated. Using a Python-based control interface, the results are dispatched to the Dobot robot arm in transformed positioning coordinates, himself allowing it to aspirate and make pick-and-places using a suction gripper. Experimental evaluation shows that robots can perform tasks mainly through the detection of reliable objects, interpret commands accurately, and reliably perform tasks. The proposed system is effective in integrating computer vision with speech interaction in the case of applications that require a human-robot interaction and use of educational robotics.

Index Terms: HSV color segmentation, homography mapping, computer vision, voice-controlled robotics, human-robot interaction, speech recognition, robotic manipulation, Dobot Magician.

I. INTRODUCTION

Robotic systems now play a much wider role in a variety of industries, including manufacturing, healthcare, logistics, and education, thanks to the quick development of intelligent automation technologies in recent years. The need for safe and intuitive interaction mechanisms has grown as robots operate more frequently in human-shared environments. The gap between human intention and robotic execution is lessened by systems that enable users to interact with robots using natural forms of communication, such as speech and visual cues. These interfaces lower the learning curve related to robotic operation and increase the usability of robotic systems for non-technical users.

Making sure the robot correctly understands and carries out user commands is a significant challenge in speech-

controlled robotic systems. Occasionally, background noise, pronunciation variations, or environmental disruptions can cause speech recognition systems to make recognition errors. Incorrect commands may cause unintentional or dangerous robot movements if robotic actions are initiated right away following speech recognition without verification. In order to increase operational reliability, it becomes imperative to incorporate an additional validation mechanism.

Vision verification is an efficient remedy to this problem. The robotic system can verify the presence of the object requested and accurately determine the location position of the object by scanning the visual surroundings prior to making a command. Object detection on the basis of color is especially appropriate in controlled settings where objects may be differentiated in terms of different color characteristics. Color segmentation (HSV) and contour detection are also image processing methods that offer computationally efficient ways of detecting objects in real-time without involving sophisticated machine learning models.

Besides object detection, there must be accurate mapping of space between the camera and the robotic workspace also to facilitate accurate robotic manipulation. Image pixel positions can be changed into real-world coordinates that are capable of describing the robot motion space by use of coordinate transformation methods like homography mapping. This conversion will make sure that the robotic arm will be able to reach the position of the detected object with fine positional accuracy.

The speech recognition, computer vision and robotic manipulation are combined to form a multi-modal control structure where each component is complementary to the other. The interface is speech-based, vision is environmental-aware and the robotic arm performs the actual task. When all these elements are incorporated into one architecture, they enhance the effectiveness of interactions and safety of operations.

The main point of the research will be to design and deploy a viable human-robot interaction system that could undertake voice-guided object manipulation operations. The system aims at identifying colored objects in a working area and allowing a robot arm to pick and place these objects according to voice commands. The proposed framework offers a relatively low-cost solution to the task by incorporating lightweight computer vision methods with speech-based interaction and are able to run in laboratory settings without any special hardware or extensive computing power.

The principal contributions of this work could be explained in the following manner:

- Creation of a robotic manipulation framework where speech is used to control the robot to select the object by giving the user a natural voice command.
- A vision based object detector module involving HSV color segmentation, contour analysis to determine colored objects in the workspace is implemented.
- Homography-based coordinate transformation used in transforming the coordinates of the camera image to robot workspace coordinates.
- Combination of speech processing, computer vision and robotic control module to allow the practice of voice-guided pick-and-place functions using the Dobot robotic arm.
- Presentation of a working human-robot interaction system that can be used as educational robotics laboratories and automation research platforms.

The rest of this paper is structured in the following way. Section II will review related literature on speech-controlled robot machines and vision-based object details detection procedures. Section III will represent the architecture of the proposed system and the interaction among the various system components. Section IV is the methodology of speech recognition, object detection, coordinate mapping and robotic control. The implementation of hardware and the experimental setup is discussed in Section V. The experimental results and the performance of the system are analyzed in Section VI. Lastly, Section VII will draw a close to the paper and provide possible research direction in the future.

II. RELATED WORK

A. Speech-Based Robotic Control Systems

Voice interaction has proved to be a viable method of enhancing communication between humans and robots. Traditional methods of controlling robots are usually through graphical interface, dedicated controllers or programming instructions, that might need technical expertise to use. Speech interfaces on the other hand allows the user to interact with the robots by speaking in natural language which makes interaction easier and robotic systems more approachable to more people.

Early voice-controlled robot applications often made use of speech recognition services on clouds. Audio signals received by a microphone in these systems were sent to remote servers to be processed in the server where the speech was transferred to advanced recognition algorithms to convert it into text commands. The accuracy of recognition through cloud-based processing was high because of the access to high-processing power and large training examples. Nevertheless, dependency on network connectivity also added some practical difficulties such as the delay in communication, the reliance on the stable internet connection, and the risk of the delay in the execution of commands.

In order to overcome such constraints, newer research has examined other methods of enhancing responsiveness and system reliability. The speech recognition technologies have progressively developed to facilitate a more efficient real-time processing and less problematic integration into robotic control systems. The current speech processing

packages enable a speech-to-text interface to be included in robots with very little computational load on a computer, permitting a natural voice interface to be used without machine-specific hardware.

Despite these advancements, many speech-controlled robotic systems still execute actions immediately after a command is recognized. Because speech recognition may occasionally produce errors due to environmental noise, accent variations, or unclear pronunciation, executing commands without verification may result in unintended robot movements. For this reason, several researchers have proposed combining speech recognition with additional sensing mechanisms that validate the command before execution. Integrating speech interfaces with perception modules such as cameras can significantly improve system safety and reliability.

B. Vision-Based Robotic Systems

Visual perception plays a key role in modern robotic systems because it allows machines to observe and understand their surroundings. By using cameras together with image processing techniques, robots can identify objects, estimate their positions, and monitor changes within the working environment. As a result, vision systems greatly contribute to improving both the accuracy and independence of robotic operations.

In the early stages of robotic vision development, most systems relied on traditional image processing approaches. Techniques such as color thresholding, edge detection, and contour extraction were widely used to detect objects in structured environments. These methods are computationally efficient and work well in situations where the workspace conditions remain stable. However, their performance can be influenced by factors such as lighting variations, background complexity, or changes in the appearance of objects.

With the advancement of artificial intelligence technologies, machine learning and deep learning approaches have been introduced to enhance robotic vision capabilities. Methods based on convolutional neural networks and other learning models can recognize more complex patterns and adapt to a wider range of environmental conditions. While these techniques can improve detection performance, they usually require large datasets and significant computational resources, which may not be necessary for applications conducted in controlled laboratory environments.

In many robotic platforms, the vision module functions independently and is mainly responsible for identifying objects or monitoring the workspace. However, when robots are controlled through human instructions—especially voice commands—visual information can also act as a verification mechanism. Without this additional confirmation, the robot may attempt to carry out an action even if the requested object is missing or incorrectly detected.

To address this issue, recent research has focused on integrating visual perception with human interaction systems. In such frameworks, the vision module verifies the presence and location of the requested object before the robot performs any physical task. The system proposed in this work follows this concept by combining camera-based object detection with speech-driven control. An overhead camera continuously observes the workspace and uses color segmentation techniques to identify objects of interest. After

confirming the object's presence and position, the robotic arm receives the corresponding instruction and performs the required pick-and-place operation.

III. PROPOSED SYSTEM ARCHITECTURE

The proposed system architecture is developed to support intuitive human-robot interaction by combining speech processing, computer vision, and robotic manipulation within a unified framework. The overall system transforms natural spoken commands into validated robotic actions through a sequence of coordinated processing steps. These steps include voice input acquisition, speech recognition, command interpretation, visual verification, coordinate transformation, and robotic execution. Each module performs a specific task while interacting with the other components to maintain reliable and safe system operation.

The interaction process begins with the voice input stage, where the user provides spoken instructions through a microphone. The microphone captures the acoustic signal and sends the audio stream to the speech processing module. At this stage, the speech recognition system analyzes the recorded signal and converts the spoken words into text. This transformation allows the robotic control program to interpret the user's instructions in a format that can be processed by the system.

Once the spoken command has been converted into text, the command interpretation module analyzes the recognized words to identify the intended robotic task. The system searches for predefined keywords that correspond to specific actions or object categories. For instance, commands such as "pick red," "pick green," or "pick blue" instruct the robotic arm to locate and manipulate objects of the corresponding color. Instead of immediately executing the command, the system introduces an additional verification stage to confirm that the requested object is present in the workspace.

This verification is carried out by the visual perception module, which uses a camera placed above the robot workspace to continuously observe the operating area. The camera captures live images that are processed using computer vision techniques implemented through the OpenCV library. Initially, each captured frame is converted from the RGB color representation to the HSV color space, which provides more reliable color detection under different lighting conditions. Color segmentation techniques are then applied to isolate objects with specific color characteristics from the background.

After potential objects are detected, contour analysis is used to determine the boundaries of each segmented region. From these contours, the system calculates the centroid of the object, which represents its location in the image frame. The centroid coordinates serve as the reference point for determining the position of the object within the camera view. Through this process, the vision module verifies both the existence and approximate location of the object before allowing the robot to move.

To enable the robot to physically reach the detected object, the system converts the image coordinates into the robot's workspace coordinates. This conversion is achieved using a homography-based mapping technique. During system calibration, corresponding points between the camera image and the robot workspace are defined to compute the homography matrix. Using this transformation, the pixel

coordinates obtained from the image are mapped into real-world coordinates that fall within the reachable workspace of the robotic arm. Additional boundary checks and calibration adjustments are applied to ensure that the calculated coordinates remain within the robot's safe operating range.

After the object location has been validated and transformed into robot coordinates, the control module generates the required motion commands for the robotic manipulator. Communication with the robot is handled through a Python-based interface that sends position commands to the robot controller. These commands define the movement sequence needed to complete the pick-and-place task.

The final stage involves the physical execution of the task using the Dobot robotic arm equipped with a suction-based gripper. The robot first moves to a position above the detected object and then lowers itself to the specified pickup height. Once the suction gripper is activated to grasp the object, the robotic arm lifts it and transports it to a predefined drop location associated with the selected object category. After placing the object at the destination, the robot returns to its standby position, allowing the system to process the next command.

By combining speech-based interaction with vision-based verification and robotic control, the proposed architecture ensures that robotic actions are executed only when both command recognition and environmental confirmation are successful. This layered verification mechanism increases system reliability and minimizes the possibility of unintended robot movements. As a result, the framework offers a practical and user-friendly solution for enabling natural voice-based control of robotic manipulation tasks.

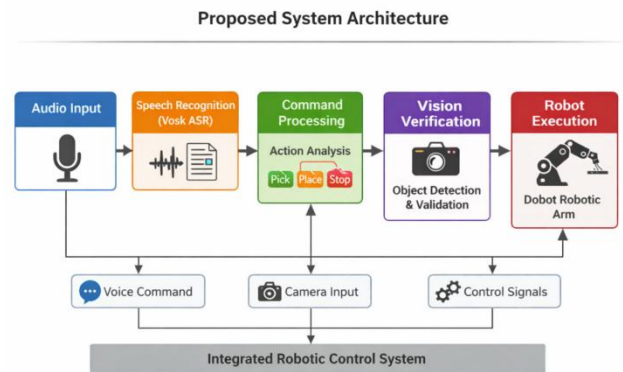


Fig:1

IV. METHODOLOGY

The proposed system operates through a structured workflow that converts spoken instructions from the user into verified robotic actions. This process is divided into several sequential stages, including voice acquisition, speech recognition, command interpretation, visual validation, coordinate transformation, and robotic execution. Each stage contributes a specific function within the processing pipeline, allowing the robot to carry out commands accurately while maintaining safe system operation.

The workflow begins with the voice input stage, where the user speaks a command through a microphone connected to the host computer. The microphone captures the acoustic

signal and converts it into a digital audio stream suitable for speech processing. This recorded audio is then forwarded to the speech recognition module, where the spoken words are analyzed and converted into text. Through this speech-to-text conversion process, the system can interpret human instructions and transform them into commands that can be understood by the robotic control software.

Once the spoken command has been converted into textual form, the command interpretation module determines the intended robotic task. The system analyzes the recognized words and compares them with a predefined set of valid commands stored in the control logic. Specific keywords related to object colors are used to identify the requested operation. For instance, commands such as “pick red,” “pick green,” or “pick blue” instruct the robotic arm to locate and manipulate objects of the corresponding color. If the recognized speech does not match the expected command patterns, the instruction is ignored to prevent unintended robot movements.

Before the robot begins any physical action, the system performs visual verification to ensure that the requested object is present in the workspace. An overhead camera continuously observes the robot’s operating area and captures live image frames. These images are processed using computer vision algorithms implemented through the OpenCV library. To improve the reliability of color detection, the captured images are first converted from the RGB color format to the HSV color space. This conversion allows color-based segmentation techniques to identify objects more effectively under varying lighting conditions.

Following the segmentation stage, contour detection is used to identify the boundaries of objects that match the specified color range. Once the relevant contour is detected, the system calculates the centroid of the object. This centroid represents the approximate location of the object within the camera frame and provides the pixel coordinates required for subsequent spatial processing.

To enable the robotic arm to physically reach the detected object, the system converts these pixel coordinates into real-world robot coordinates. This transformation is performed using a homography-based mapping approach obtained during system calibration. The homography matrix establishes a relationship between the camera image plane and the robot workspace, allowing the system to accurately translate image positions into coordinates that the robot can reach. Additional safety checks and calibration offsets are applied to ensure that the computed positions remain within the robot’s allowable movement range.

After the object location has been successfully mapped to the robot coordinate system, the control module generates the necessary motion commands for the robotic arm. These commands are transmitted through a Python-based interface that communicates directly with the robot controller. The robot then follows a predefined sequence of movements designed for object manipulation. First, the robotic arm moves to a position above the detected object and then lowers itself vertically to the picking height.

Once the robot reaches the designated pickup point, the suction-based gripper is activated to grasp the object. The robotic arm then lifts the object to a safe height and transports it toward a predefined drop location associated with the selected object category. After releasing the object

at the destination, the robot returns to its standby position, preparing the system for the next user command.

The modular design of the proposed methodology supports efficient real-time operation while maintaining system reliability and safety. By integrating speech-based command input, computer vision-based object detection, coordinate mapping, and robotic manipulation, the system ensures that actions are executed only when both command interpretation and visual verification conditions are satisfied. This integrated framework provides a practical solution for implementing intuitive voice-controlled robotic systems in laboratory environments and educational robotics applications.

Methodology Flow of the Proposed System

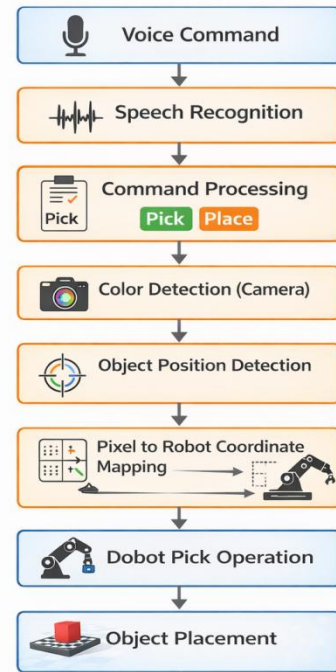


Fig:2

V. HARDWARE IMPLEMENTATION

The experimental setup for the proposed system consists of a Dobot Magician robotic arm connected to a desktop computer, along with a USB microphone and an overhead camera used for visual monitoring. The robotic arm acts as the main actuation device responsible for performing pick-and-place operations within a defined workspace. Communication between the computer and the robot is established through a USB serial interface, allowing the control program to send movement commands to the robot in real time using a Python-based control library.

The Dobot Magician provides four degrees of freedom, enabling movement along the X, Y, and Z axes as well as rotational control. These motion capabilities allow the robot to accurately reach and manipulate objects within its workspace. Before conducting the experiments, the robot was calibrated manually to determine important reference coordinates required for reliable operation. These reference points include the home position, approach positions above the object, the contact point used for picking the object, and

predefined drop locations. Defining these coordinates helps ensure that the robot performs consistent and repeatable movements during multiple pick-and-place cycles. A vacuum-based suction gripper is attached as the robot's end effector, enabling it to securely pick lightweight objects during the manipulation process.

User commands are provided through voice input using a USB microphone connected to the host computer. The microphone captures spoken instructions with an appropriate sampling rate to maintain clear audio quality. The recorded audio signal is then processed by the speech recognition module, which converts the spoken commands into text. These textual commands are interpreted by the robotic control program, allowing users to interact with the robotic system using simple voice instructions instead of traditional manual controls.

Along with speech input, the system incorporates a vision module that monitors the robot workspace and verifies object presence before executing robotic movements. A high-resolution camera is mounted above the workspace to provide a top-view perspective of the operating area. The captured frames are processed using computer vision techniques implemented through the OpenCV library. To improve the accuracy of color detection under different lighting conditions, each image is converted from the RGB color representation to the HSV color space. This conversion allows the system to apply color thresholds more effectively in order to identify objects of specific colors within the workspace.

After the color segmentation process, contour detection algorithms are used to identify the boundaries of the detected objects. From these contours, the centroid of the object is calculated to estimate its position in the camera image. The centroid coordinates serve as an approximate location of the object and are used to determine where the robotic arm should move to pick the object.

To allow the robotic arm to reach the detected object accurately, the system converts the pixel coordinates from the camera image into physical coordinates within the robot workspace. This transformation is carried out using a homography-based mapping method obtained during system calibration. The homography transformation establishes a relationship between the camera view and the robot workspace, enabling accurate conversion of image coordinates into real-world positions that the robot can reach. Additional safety constraints are applied to ensure that the calculated coordinates remain within the robot's allowable operating range.

The entire system operates through coordinated interaction between the speech recognition module, the computer vision module, and the robotic control module. The software implementation was developed using the Python programming language, which allows flexible integration of the different components. Libraries such as OpenCV are used for image processing, while robot communication libraries enable motion control of the Dobot manipulator.

To evaluate the system, a series of experiments were conducted in a controlled laboratory environment. Colored objects were placed at different locations within the robot workspace, and voice commands were issued to instruct the robot to pick and place objects based on their color. Multiple trials were performed to evaluate the reliability of the speech

recognition module, the accuracy of the vision-based object detection system, and the precision of the robotic manipulation process.

The experimental observations showed that the system operated reliably with effective coordination between all modules. The vision system was able to identify colored objects accurately, while the speech interface allowed users to provide commands in a simple and intuitive way. In addition, combining speech commands with visual confirmation significantly reduced the chances of incorrect robot actions caused by misinterpreted voice inputs.

Overall, the experimental results demonstrate the practicality of the proposed human-centric robotic control system. By integrating speech interaction, computer vision, and robotic manipulation, the framework provides an efficient approach for enabling natural human-robot collaboration. Such systems can be particularly useful in educational robotics laboratories, training environments, and research platforms focused on developing intelligent robotic interaction technologies.



Fig:3

VI. RESULTS AND DISCUSSION

The performance of the proposed human-robot interaction system was evaluated through a series of experiments conducted in a controlled laboratory environment. The evaluation focused on three main factors: the accuracy of speech command recognition, the overall response time of the system, and the reliability of the robotic task execution. During the testing phase, users repeatedly issued voice commands such as pick red, pick green, pick blue, and stop while the system operated under typical indoor conditions. These experiments were designed to determine how effectively the system could interpret spoken instructions and convert them into appropriate robotic actions.

The experimental observations showed that the speech recognition module operated reliably with minimal processing delay. On average, the time between issuing a voice command and the start of robotic movement ranged between 1 and 1.5 seconds. This response time includes several processing steps such as capturing the audio signal, converting speech into text, interpreting the command, verifying object presence through the vision module, mapping the coordinates, and sending motion instructions to the robot. Since speech processing was carried out locally on the host computer, the system avoided delays associated with cloud-based processing, resulting in faster command execution.

Speech recognition accuracy was also examined through repeated trials. When users spoke clearly and background

noise remained moderate, the system achieved a recognition accuracy of approximately 90–95%. A slight reduction in performance was observed when environmental noise increased or when overlapping speech occurred. However, microphone calibration during system initialization helped reduce the effect of noise and improved the overall stability of speech acquisition.

The vision module played an important role in maintaining the safety and correctness of the robotic operation. Before initiating any robotic movement, the system confirmed the presence and approximate location of the target object using camera-based color detection. The computer vision algorithm, which uses HSV color segmentation combined with contour detection, successfully identified colored objects placed within the workspace. This verification stage ensured that the robot only executed commands when the requested object was actually present, thereby preventing errors that might occur due to incorrect speech recognition.

The coordinate transformation mechanism was also tested to evaluate its ability to accurately convert image coordinates into robot workspace coordinates. A homography-based mapping method was used for this purpose. Experimental results showed that this transformation provided a reliable relationship between the camera view and the robot's operating plane. As a result, the robotic arm was able to approach the detected object positions with only minimal positional error.

During repeated hardware experiments, the Dobot robotic arm demonstrated stable and consistent motion control. The robot executed the pick-and-place sequence by moving to an approach position above the object, descending to the pickup location, activating the suction-based gripper, and then transporting the object to the designated drop point. The vacuum gripper maintained a firm hold on the objects throughout the trials, and no noticeable slipping was observed during the manipulation process.

Another key observation was the effective synchronization between the different modules of the system. The speech recognition, vision processing, and robotic control components worked together in a coordinated manner. This ensured that robotic actions were triggered only after both command recognition and object verification were completed successfully. The modular design also allowed each subsystem to operate independently while maintaining smooth communication with the other modules.

Overall, the experimental results demonstrate that the proposed framework can successfully support real-time robotic control through voice interaction combined with visual verification. The integration of speech commands, camera-based object detection, and accurate robotic manipulation enhances both system reliability and operational safety. These findings indicate that the developed system is well suited for applications in educational robotics laboratories, training environments, and research platforms focused on human–robot interaction.

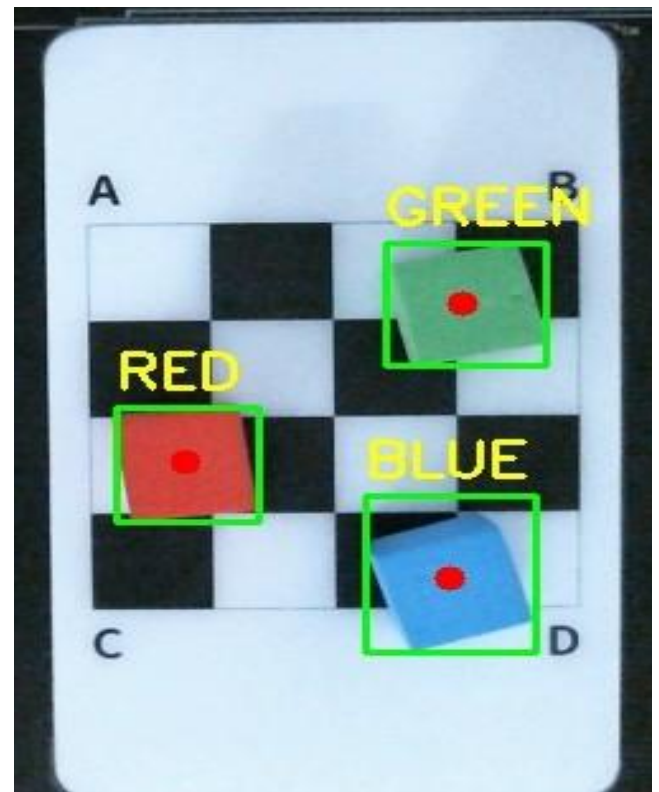


Fig:4a



Fig:4b



Fig:4c

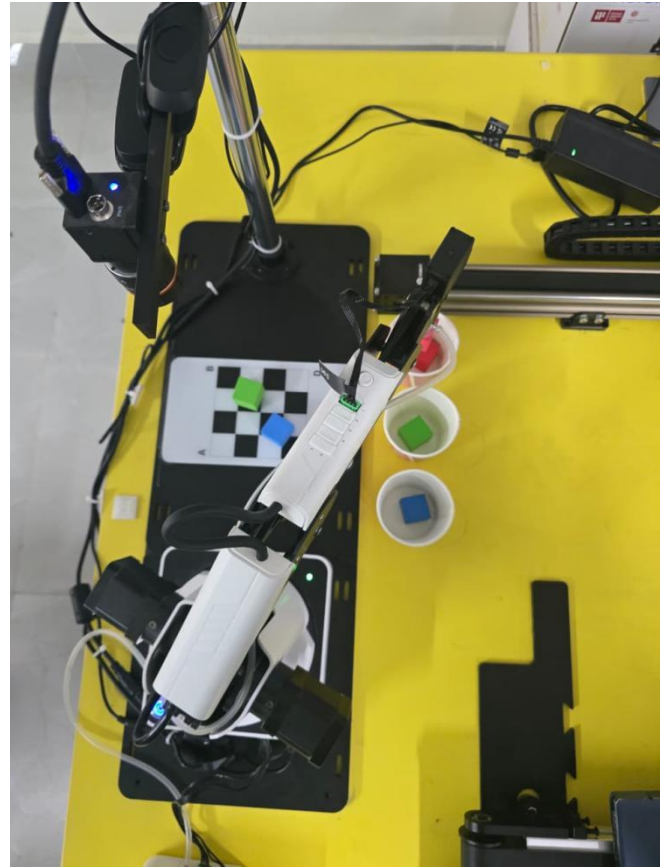


Fig:4e



Fig:4d

VII. CONCLUSION

This study introduced a human-centered robotic control framework that combines speech interaction with vision-based object verification to enable reliable robotic task execution. The proposed system allows users to operate a Dobot robotic arm through simple voice commands, removing the need for complex programming or traditional control interfaces. By enabling natural speech as the primary input method, the system makes robotic interaction more intuitive and accessible, particularly in environments where ease of use is important.

To improve the safety and reliability of the system, a computer vision module was incorporated into the overall architecture. This module continuously observes the robot workspace using a camera and processes the captured images to confirm the presence and position of the requested object before any robotic motion is initiated. This verification step helps prevent errors that could occur due to incorrectly recognized voice commands or the absence of the intended object in the workspace. By combining speech recognition, visual perception, and robotic manipulation, the system establishes a more dependable control approach compared with voice-only robotic systems.

The experimental results demonstrated that the system performs effectively under controlled laboratory conditions. The speech recognition module showed stable command detection with minimal processing delay, while the Dobot robotic arm successfully completed pick-and-place operations using predefined coordinate positions. Furthermore, incorporating visual verification increased the robustness of the system by ensuring that robotic actions

were executed only when the target object was properly detected and located within the workspace.

The proposed framework provides a practical and cost-effective solution for developing intuitive robotic control systems. Such an approach can be valuable in educational robotics laboratories, industrial training environments, and research platforms that focus on human–robot collaboration. In addition, the modular architecture of the system makes it easier to expand or integrate with other sensing and control technologies in future implementations.

Future work may focus on improving the vision module by incorporating more advanced object detection techniques. The use of modern computer vision methods or learning-based models could allow the system to recognize a broader range of objects and operate effectively under more complex environmental conditions. Further enhancements may also include supporting continuous voice interaction, improving natural language understanding, and integrating the system with advanced robotic middleware platforms. These developments would enhance the intelligence, flexibility, and real-world applicability of the proposed human–robot interaction framework.

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